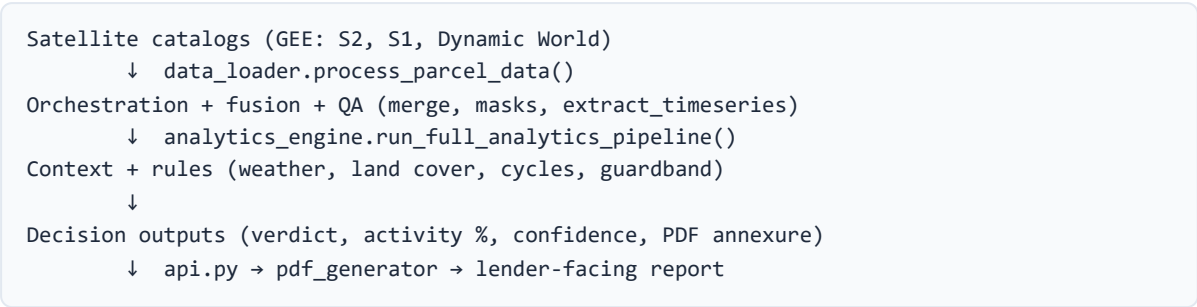


AdvaRisk & the EO Downstream Thesis

Technical addendum — how industry thinking (SatSure / EO application layer) maps to this codebase ·
24 May 2026, 12:49 AM

Core thesis (EO industry): More satellites do not automatically improve decisions. The binding constraint is turning pixels into answers that fit *how a customer decides* — orchestration, context, and decision-ready modelling. AdvaRisk is built in that direction, not as a data-vendor stack.

1. WHERE VALUE IS CREATED



Upstream (~20% of product code): fetch indices via Earth Engine. **Downstream** (~80%): smoothing, empirical logic, seasonal tables, environmental veto, PDF/API packaging.

2. THE THREE “HARD PARTS” — MAPPED TO FILES

Layer	What it means	AdvaRisk today
1. Orchestration	Reliable multi-source pipelines, lineage, automated QA	<code>api.py</code> parallel workers (satellite, location, weather); <code>data_loader.py</code> GEE init, S2/S1/DW collections, <code>extract_timeseries</code> , CSV/pickle; replay via GCS pickle.
2. Context	Fuse EO with weather, land semantics, sector meaning	Dynamic World 9-class fusion; <code>rain_temp.py</code> in PDF; parcel KML + static map; lender lens : same NDVI read through crop-probability + trees/water guardband.
3. Modelling	Outputs usable in workflow; speed vs accuracy trade-offs	<code>detect_crop_cycles</code> , <code>apply_empirical_logic</code> ; Whittaker daily series; verdict: <code>activity_ratio > 15%</code> → “Crop activity detected”.

3. PARALLEL TO AG CREDIT (SATSOURCE PATTERN)

Mature EO-credit product

AdvaRisk (this repo)

- Many parameters + calibrated score
- Embedded in business rule engine / LOS
- Weeks → minutes; fewer field visits
- Indices + land cover + seasons + cycles + confidence
- API verdict + intelligence PDF (not yet LOS)
- Automated parcel pipeline; threshold-based verdict

Same **decision family** (is this land actively used for agriculture / what is the risk signal?), earlier maturity: strong on *signal assembly*, lighter on *score calibration with banks and enterprise integration*.

4. TRADE-OFFS WE ALREADY ENCODE

Trade-off	Code choice	Effect
Speed vs density	<code>limit_to_60</code> on S2	Cheaper runs; fewer optical scene dates
Readability vs fidelity	Whittaker + daily <code>resample('D')</code>	Smooth reports; monsoon can look “continuous”
Recall vs lender caution	Environmental guardband	Suppresses crop calls when trees/water dominate
Transparency	<code>raw_indices_df</code> scatter vs smoothed lines	Honest scene dates — if explained (see companion PDF)

5. GAPS VS “ESSENTIAL WORKFLOW” (PILOT → PRODUCTION)

- **QA:** no minimum clear-pixel fraction per parcel before accepting monsoon NDVI scenes.
- **Semantics:** 15% activity threshold not yet co-defined with risk teams as score bands.
- **Integration:** PDF/API output, not wired into loan origination or audit trail standards.
- **Context breadth:** no socio-economic / ground-truth layers in pipeline yet.

6. WHERE THE MOAT SHOULD SIT (FOR THIS TEAM)

Not the moat: downloading more Sentinel scenes.

The moat: (1) trusted orchestration + QA, (2) sector-specific fusion and rules banks accept, (3) embedding in their decision engine so switching cost is high. Compounding comes from each deployment improving pipelines and the next sector shipping faster.

Answer to “data vs models vs decision integration?” — For AdvaRisk, lasting advantage aligns with **decision integration + contextual fusion**; satellite data and heuristics/models are enablers. Next engineering priority: tighten downstream honesty (cloud QA, optical vs SAR separation) and co-design verdict/score meaning with customers — not add indices alone.